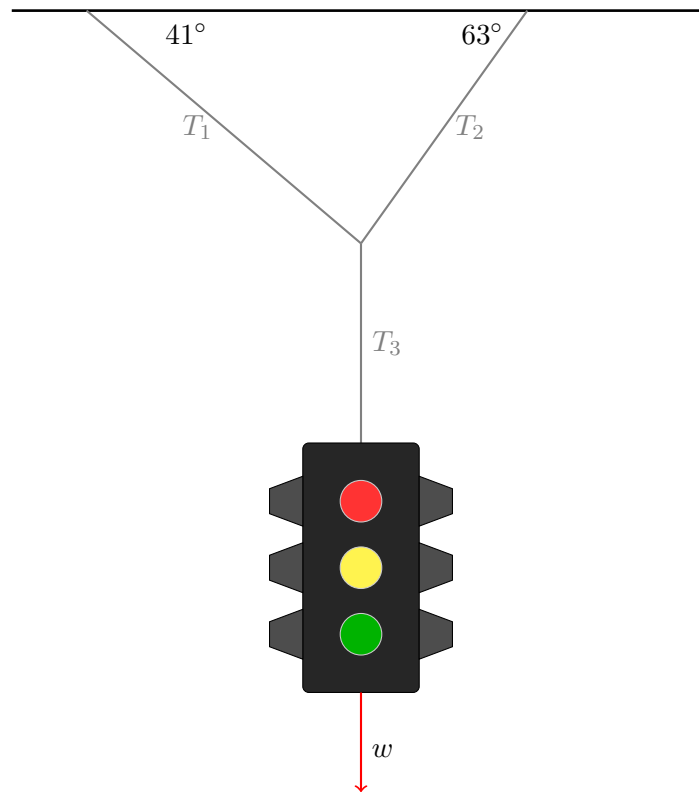


Problem 1. Find the tension in each of the three cables supporting the traffic light if it weighs 175 N.



Solution.

Since the traffic light is in equilibrium, the net force on the light and on the junction of the cables must be zero.

First, consider the vertical cable. Since it directly supports the traffic light,

$$T_3 = W = 175 \text{ N}$$

Now analyze the junction where the three cables meet. Take the angles of 41° and 63° with respect to the horizontal.

For equilibrium in the horizontal direction,

$$T_2 \cos(63^\circ) = T_1 \cos(41^\circ)$$

So,

$$T_2 = \frac{T_1 \cos(41^\circ)}{\cos(63^\circ)}$$

For equilibrium in the vertical direction,

$$T_1 \sin(41^\circ) + T_2 \sin(63^\circ) = T_3$$

Substitute $T_3 = 175$ N and the expression for T_2 :

$$T_1 \sin(41^\circ) + \left(\frac{T_1 \cos(41^\circ)}{\cos(63^\circ)} \right) \sin(63^\circ) = 175$$

Factor out T_1 :

$$T_1 \left[\sin(41^\circ) + \frac{\cos(41^\circ) \sin(63^\circ)}{\cos(63^\circ)} \right] = 175$$

Now substitute numerical values:

$$T_1 \left[0.656 + \frac{(0.755)(0.891)}{0.454} \right] = 175$$

$$T_1(2.14) \approx 175$$

Thus,

$$T_1 \approx 81.8 \text{ N}$$

Now solve for T_2 :

$$T_2 = \frac{(81.8)(\cos 41^\circ)}{\cos 63^\circ}$$

$$= \frac{(81.8)(0.755)}{0.454}$$

$$\approx 136 \text{ N}$$

Therefore,

$$T_1 \approx 81.8 \text{ N}$$

$$T_2 \approx 136 \text{ N}$$

$$T_3 = 175 \text{ N}$$

Problem 4. A 2.25-kg fireworks shell is fired straight up from a mortar and reaches a height of 111 m. (a) Neglecting air resistance, calculate the shell's velocity when it leaves the mortar. (b) The mortar itself is a tube 0.450 m long. Calculate the average acceleration of the shell in the tube as it goes from zero to the velocity found in part (a). (c) What is the average force on the shell in the mortar (as measured in units of its own weight)?

Solution.

At the top of the motion, the shell has velocity $v = 0$. Using

$$v^2 = v_0^2 + 2a\Delta y$$

with $a = -g$ and $\Delta y = 111$ m, we get

$$\begin{aligned} 0 &= v_0^2 - 2g(111) \\ v_0^2 &= 2g(111) \\ v_0 &= \sqrt{2(9.8)(111)} \\ v_0 &\approx 46.6 \text{ m/s} \end{aligned}$$

(a) Thus, the shell leaves the mortar with speed

$$v_0 \approx 46.6 \text{ m/s}$$

Now use the motion inside the mortar tube. The shell starts from rest and travels 0.450 m while reaching speed 46.6 m/s. Using

$$v^2 = v_i^2 + 2a\Delta x$$

with $v_i = 0$:

$$\begin{aligned} (46.6)^2 &= 2a(0.450) \\ a &= \frac{(46.6)^2}{0.900} \\ a &\approx 2.41 \times 10^3 \text{ m/s}^2 \end{aligned}$$

(b) So the average acceleration in the tube is

$$a \approx 2.41 \times 10^3 \text{ m/s}^2$$

To find the average force exerted on the shell by the mortar, apply Newton's second law:

$$F_{\text{mortar}} - mg = ma$$

so

$$\begin{aligned} F_{\text{mortar}} &= m(a + g) \\ &= (2.25)(2.41 \times 10^3 + 9.8) \\ &\approx 5.46 \times 10^3 \text{ N} \end{aligned}$$

The shell's weight is

$$W = mg = (2.25)(9.8) = 22.05 \text{ N}$$

(c) Thus, in units of its own weight,

$$\begin{aligned} \frac{F_{\text{mortar}}}{W} &= \frac{5.46 \times 10^3}{22.05} \\ &\approx 248 \end{aligned}$$

Therefore,

$$\begin{aligned} v_0 &\approx 46.6 \text{ m/s} \\ a &\approx 2.41 \times 10^3 \text{ m/s}^2 \\ F_{\text{mortar}} &\approx 248 \text{ times its own weight} \end{aligned}$$

Problem 5. A service elevator takes a load of garbage, mass 10 kg, from a floor of a skyscraper under construction, down to ground level, accelerating downward at a rate of 1.2 m/s^2 . Find the magnitude of the force the garbage exerts on the floor of the service elevator.

Solution.

The net force acting on the garbage is given by $F_{\text{net}} = F - mg = -ma$.

Solving for F :

$$F = mg - ma = m(g - a) = (10 \text{ kg})(9.8 \frac{\text{m}}{\text{s}^2} - 1.2 \frac{\text{m}}{\text{s}^2})$$
$$F = 86 \text{ N}$$

Problem 6. Two blocks are connected by a massless rope as shown below. The mass of the block on the table is 5.0 kg and the hanging mass is 0.9 kg. The table is frictionless, and the pulley is frictionless and massless. (a) Find the acceleration of the blocks. (b) Find the tension in the rope. (c) Find the speed with which the hanging mass hits the floor if it starts from rest and is initially located 0.8 m from the floor.

Solution.

Let $m_1 = 5.0 \text{ kg}$ (block on table) and $m_2 = 0.9 \text{ kg}$ (hanging mass).

For the system, the net external force is due to the weight of the hanging mass:

$$F_{\text{net}} = m_2 g$$

Applying Newton's second law to the system:

$$m_2 g = (m_1 + m_2) a$$
$$a = \frac{m_2 g}{m_1 + m_2} = \frac{(0.9)(9.8)}{5.0 + 0.9}$$
$$a \approx 1.49 \text{ m/s}^2$$

(a) $a \approx 1.5 \text{ m/s}^2$

Now solve for the tension using m_1 :

$$T = m_1 a = (5.0)(1.49)$$
$$T \approx 7.45 \text{ N}$$

(b) $T \approx 7.5 \text{ N}$

For the speed, use kinematics:

$$v^2 = v_0^2 + 2a\Delta y$$

Since $v_0 = 0$ and $\Delta y = 0.8$ m:

$$v^2 = 2(1.49)(0.8)$$

$$v^2 \approx 2.38$$

$$v \approx \sqrt{2.38} \approx 1.54 \text{ m/s}$$

(c) $v \approx 1.5$ m/s

Problem 7. Suppose you have a 129-kg wooden crate resting on a wood floor, with coefficient of static friction 0.55 between these wood surfaces. (a) What maximum force can you exert horizontally on the crate without moving it? (b) If you continue to exert this force once the crate starts to slip, what will its acceleration then be? The coefficient of sliding friction is known to be 0.33 for this situation.

Solution.

The maximum static friction force is

$$f_{s,\max} = \mu_s N$$

Since the force is applied horizontally, the normal force is

$$N = mg$$

So,

$$\begin{aligned} f_{s,\max} &= \mu_s mg \\ &= (0.55)(129 \text{ kg})(9.8 \text{ m/s}^2) \\ &\approx 695.3 \text{ N} \end{aligned}$$

Thus, the maximum force that can be exerted without moving the crate is

$$F \approx 695 \text{ N}$$

Once the crate starts to slip, the friction force becomes kinetic:

$$f_k = \mu_k N = \mu_k mg$$

So,

$$\begin{aligned} f_k &= (0.33)(129 \text{ kg})(9.8 \text{ m/s}^2) \\ &\approx 417.2 \text{ N} \end{aligned}$$

If you continue to exert the same force as before, then the net force is

$$\begin{aligned} F_{\text{net}} &= F - f_k \\ &= 695.3 \text{ N} - 417.2 \text{ N} \\ &\approx 278.1 \text{ N} \end{aligned}$$

Using Newton's second law,

$$\begin{aligned} a &= \frac{F_{\text{net}}}{m} \\ &= \frac{278.1 \text{ N}}{129 \text{ kg}} \\ &\approx 2.16 \text{ m/s}^2 \end{aligned}$$

Therefore,

$$\begin{aligned} F &\approx 695 \text{ N} \\ a &\approx 2.16 \text{ m/s}^2 \end{aligned}$$

Problem 11. A car rounds an unbanked curve of radius 60 m. If the coefficient of static friction between the road and car is 0.7, what is the maximum speed at which the car traverses the curve without slipping?

Solution.

For an unbanked curve, static friction provides the centripetal force:

$$f_s = \frac{mv^2}{r}$$

The maximum friction force is

$$f_{s,\text{max}} = \mu_s N = \mu_s mg$$

Set these equal at the threshold of slipping:

$$\mu_s mg = \frac{mv^2}{r}$$

Cancel m and solve for v :

$$\begin{aligned} v^2 &= \mu_s gr \\ v &= \sqrt{\mu_s gr} \end{aligned}$$

Substitute values:

$$\begin{aligned} v &= \sqrt{(0.7)(9.8)(60)} \\ v &= \sqrt{411.6} \\ v &\approx 20.3 \text{ m/s} \end{aligned}$$

Therefore,

$$v \approx 20.3 \text{ m/s}$$

Problem 13. A helicopter with mass 2.85×10^4 kg has a position given by

$$\vec{r}(t) = (0.020t^3)\hat{i} + (2.2t)\hat{j} - (0.060t^2)\hat{k} \text{ m.}$$

Find the net force on the helicopter at $t = 3.2$ s.

Solution.

First find the acceleration by differentiating the position twice.

Velocity:

$$\vec{v}(t) = \frac{d\vec{r}}{dt} = (0.060t^2)\hat{i} + (2.2)\hat{j} - (0.120t)\hat{k}$$

Acceleration:

$$\vec{a}(t) = \frac{d\vec{v}}{dt} = (0.120t)\hat{i} + 0\hat{j} - (0.120)\hat{k}$$

Evaluate at $t = 3.2$ s:

$$\begin{aligned}\vec{a}(3.2) &= (0.120)(3.2)\hat{i} + 0\hat{j} - 0.120\hat{k} \\ &= (0.384)\hat{i} - (0.120)\hat{k} \text{ m/s}^2\end{aligned}$$

Now use Newton's second law $\vec{F} = m\vec{a}$:

$$\begin{aligned}\vec{F} &= (2.85 \times 10^4) \left[(0.384)\hat{i} - (0.120)\hat{k} \right] \\ &= (1.09 \times 10^4)\hat{i} - (3.42 \times 10^3)\hat{k} \text{ N}\end{aligned}$$

Therefore,

$$\vec{F} \approx (1.09 \times 10^4)\hat{i} - (3.42 \times 10^3)\hat{k} \text{ N}$$

Problem 14. A student is attempting to move a 34-kg mini-fridge into her dorm room. During a moment of inattention, the mini-fridge slides down a 35° incline at constant speed when she applies a force of 27 N acting up and parallel to the incline. What is the coefficient of kinetic friction between the fridge and the surface of the incline?

Solution.

Since the fridge moves at constant speed, the net force along the incline is zero:

$$\sum F = 0$$

Forces along the incline:

- Component of gravity down the incline: $mg \sin \theta$
- Applied force up the incline: F
- Kinetic friction up the incline: $f_k = \mu_k N$

Thus,

$$mg \sin \theta = F + f_k$$

The normal force is

$$N = mg \cos \theta$$

So,

$$mg \sin \theta = F + \mu_k mg \cos \theta$$

Solve for μ_k :

$$\mu_k = \frac{mg \sin \theta - F}{mg \cos \theta}$$

Substitute values:

$$\begin{aligned}\mu_k &= \frac{(34)(9.8) \sin(35^\circ) - 27}{(34)(9.8) \cos(35^\circ)} \\ \mu_k &= \frac{(333.2)(0.574) - 27}{(333.2)(0.819)} \\ \mu_k &= \frac{191.3 - 27}{272.9} \\ \mu_k &\approx \frac{164.3}{272.9} \approx 0.60\end{aligned}$$

Therefore,

$$\mu_k \approx 0.60$$

Problem 15. A stunt cyclist rides on the interior of a cylinder 12 m in radius. The coefficient of static friction between the tires and the wall is 0.5. Find the value of the minimum speed for the cyclist to perform the stunt.

Solution.

For this motion, the normal force provides the centripetal force:

$$N = \frac{mv^2}{r}$$

Static friction acts upward to balance the weight:

$$f_s = mg$$

At the minimum speed, friction is at its maximum:

$$f_{s,\max} = \mu_s N$$

So,

$$mg = \mu_s N$$
$$mg = \mu_s \frac{mv^2}{r}$$

Cancel m and solve for v :

$$g = \mu_s \frac{v^2}{r}$$
$$v^2 = \frac{gr}{\mu_s}$$
$$v = \sqrt{\frac{gr}{\mu_s}}$$

Substitute values:

$$v = \sqrt{\frac{(9.8)(12)}{0.5}}$$
$$v = \sqrt{235.2}$$
$$v \approx 15.3 \text{ m/s}$$

Therefore,

$$v \approx 15.3 \text{ m/s}$$